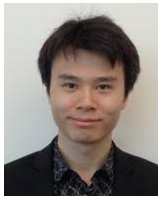
Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements



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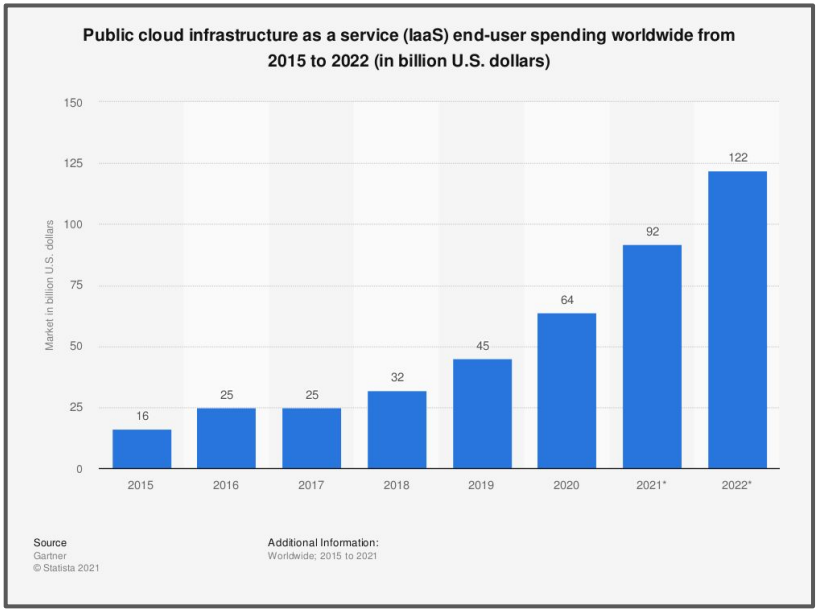
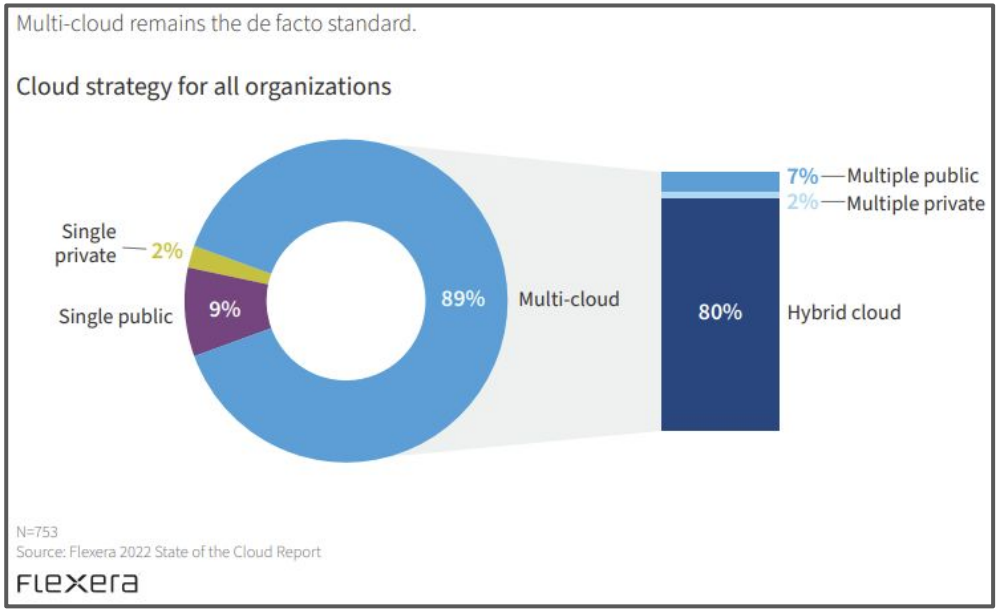
Market of Infrastructure-as-a-Service

- Scalable computing, storage, and networking resources on demand.



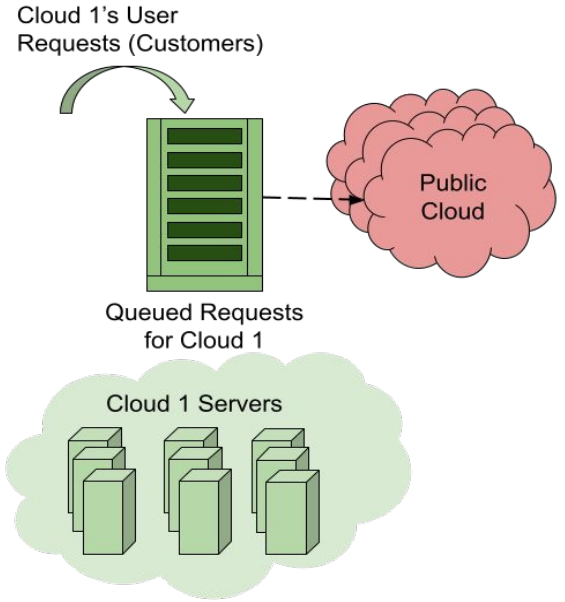


Hybrid Cloud

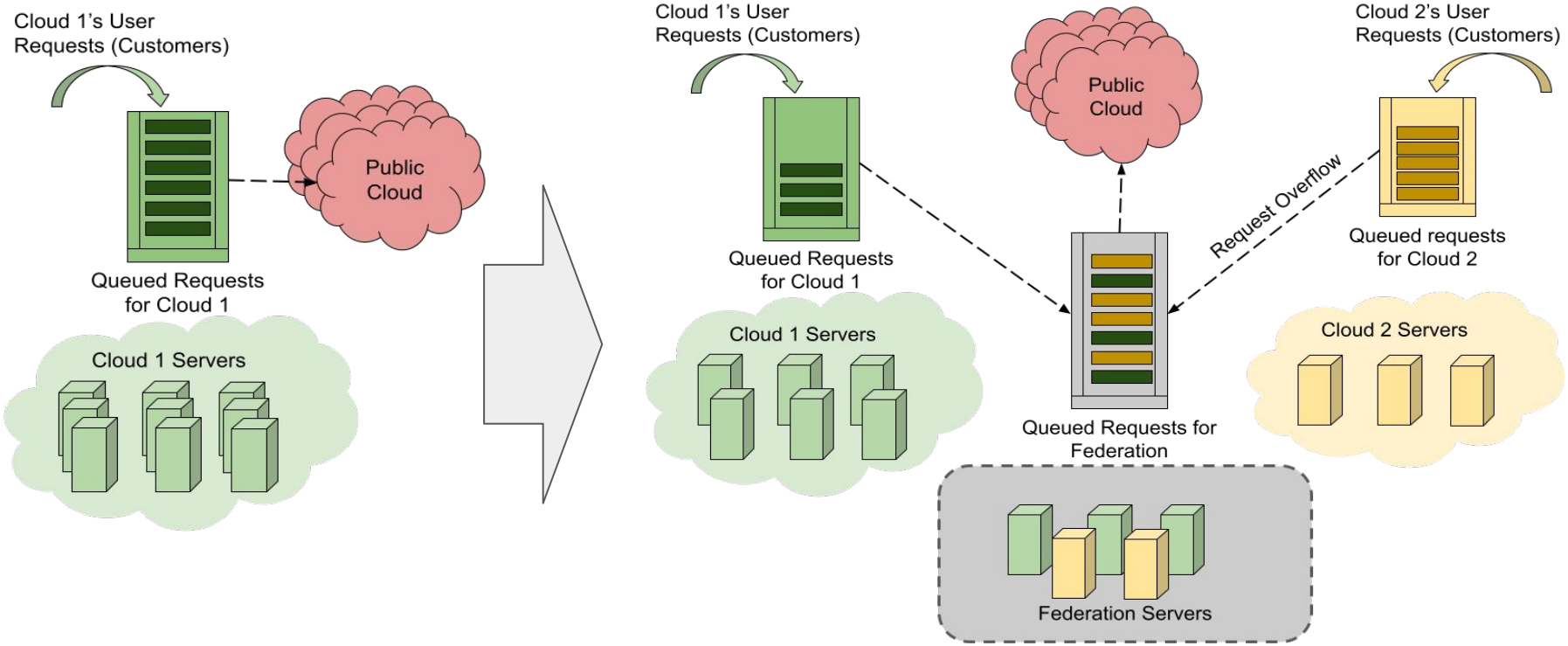
- An **under-provisioned** cloud can forward requests to public clouds
- Address **variable workloads** and **peak loads**

Does not help

- Over-provisioned clouds
- Long-term workload increase



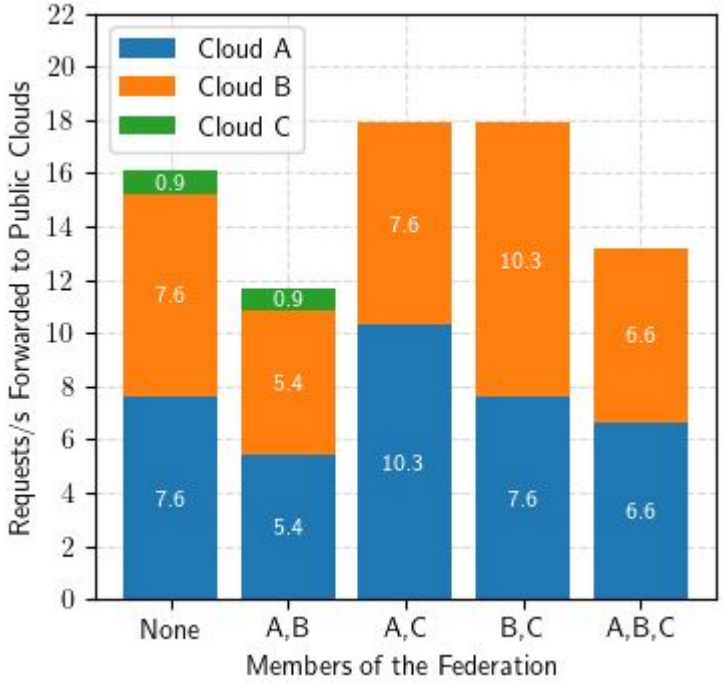
A Federation of Hybrid Clouds



Motivating Example

Under the **same arrival rate** (100) and **number of servers** (100), each cloud has individual **latency requirement** (QoS) in unit of mean service time

- Cloud A: 0 (no wait-time)
- Cloud B: 0 (no wait-time)
- Cloud C: 1 (wait upto 1 request)



Compare to state-of-the-art

Maximizing **performance** by **searching** for the best capacity or workload sharing strategy

- Darzanos 2019, Li 2021: model each cloud as a M/M/1 queue to predict performance.
- Chen 2017, Ray 2021: Invent heuristics to search for nash-stability to satisfy QoS.

Maximizing **Profit** and promote **stability** of a federation

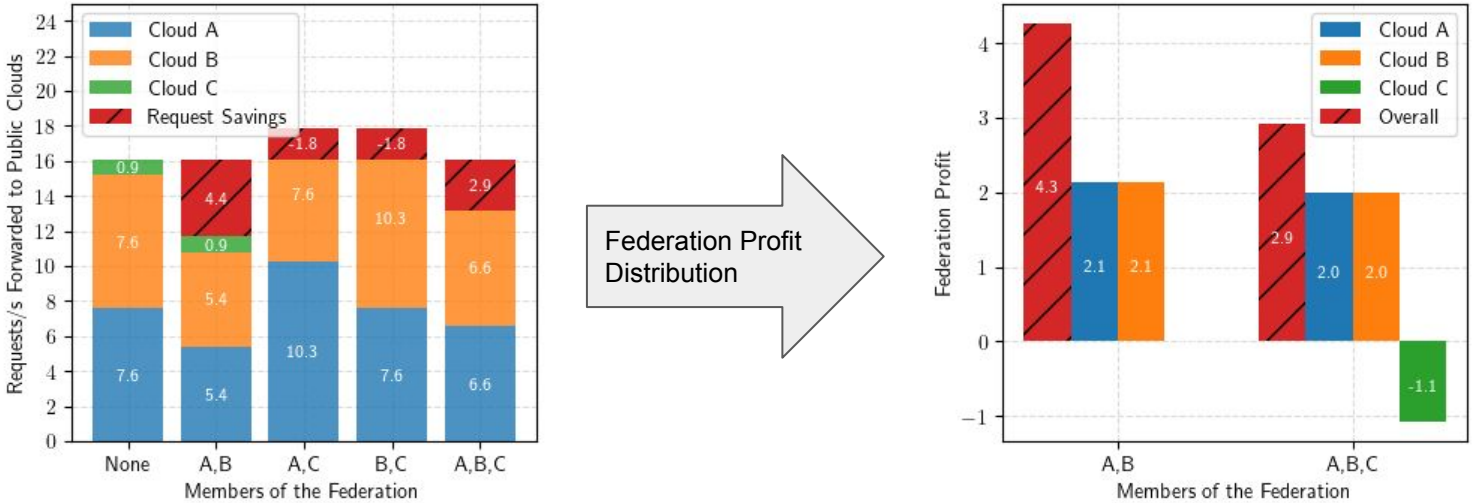
- Lin 2019, Hammoud 2020: Play **repeated game** and **evolutionary game** to let each cloud member adjust sharing strategy.

Maintaining **QoS** (maximum average wait-time)

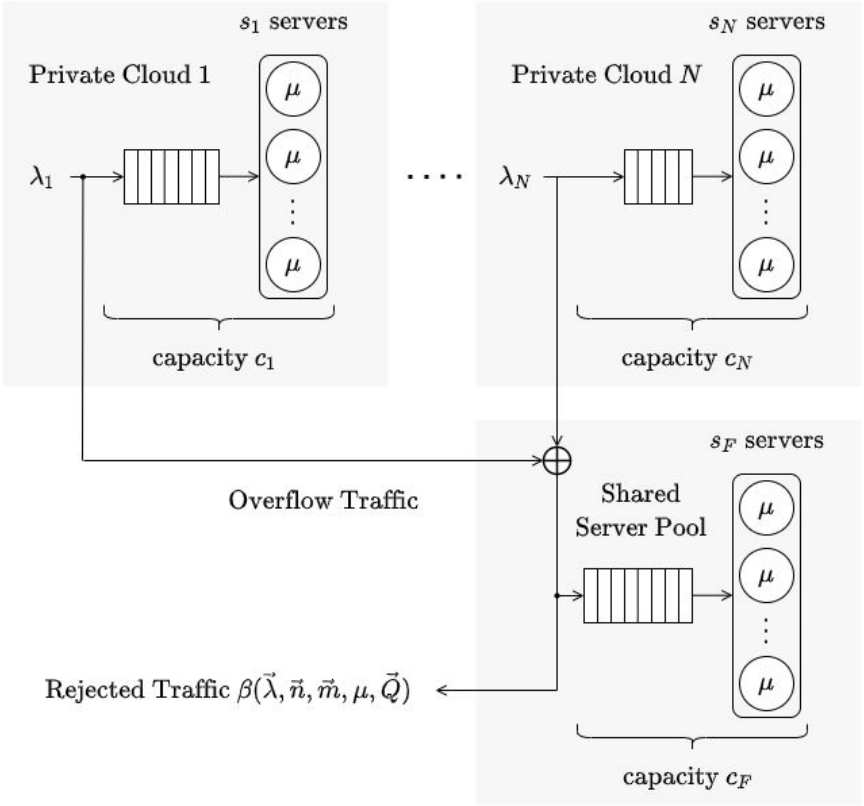
- Ray 2018, Darzanos 2019: **penalize violation** and only consider **homogeneous** QoS.

Main question

- What is the **optimal sharing strategy** when forming a federation?
- Any reason to **not form** a federation with certain clouds?
- How to **distribute** federation **revenue** according to performance?

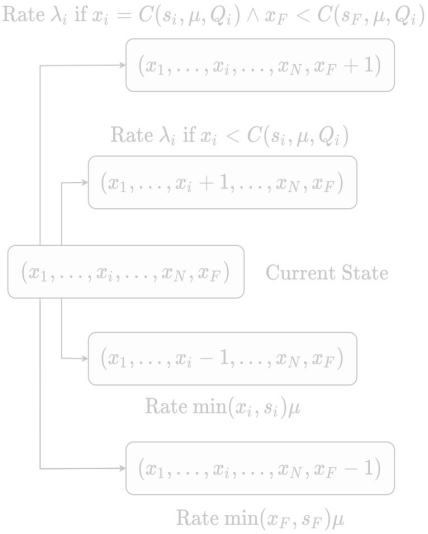


Performance Prediction



For each cloud i:

- λ_i = arrival rate
- μ = service rate (globally homogeneous)
- n_i = total number of servers
- m_i = number of servers shared to the federation
- Q_i = latency requirement (QoS) in unit of mean service time

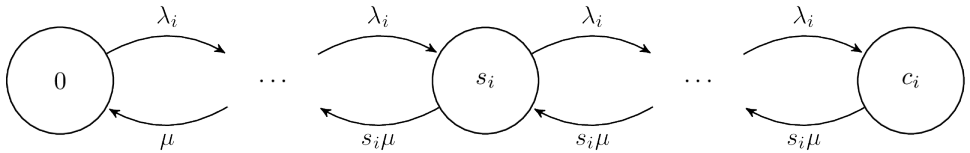


QoS and Capacity

$C(s, \mu, Q) = s + q = s + \lfloor \mu Q s \rfloor$

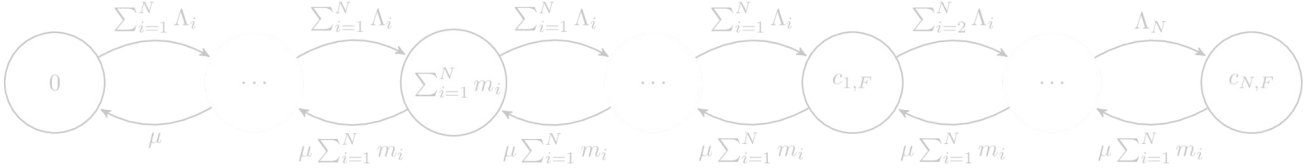
For each cloud i:

- $s_i = n_i - m_i$ = number of servers available
- c_i = capacity = $\lfloor (1 + Q_i)\mu s_i \rfloor$

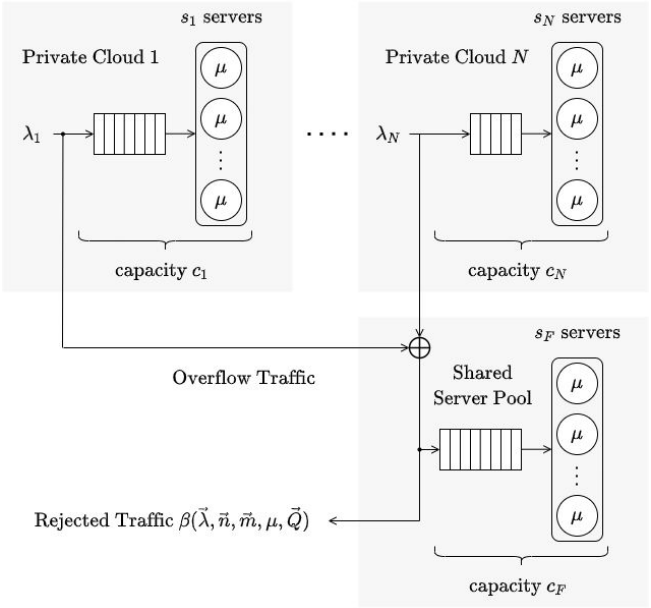


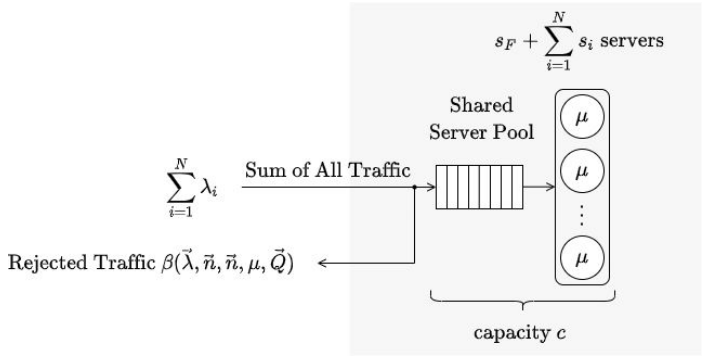
For the federation:

- Λ_i = overflow from cloud i
- $c_{i,F}$ = capacity till rejecting i's overflow = $\lfloor (1 + Q_i)\mu \sum_{i=1}^{N} m_i \rfloor$
- $c_{1,F} \leq c_{2,F} \leq ... \leq c_{N,F}$
- if $Q_i = Q \; \forall i \in \{1, ..., N\} \rightarrow c_{1,F} = c_{2,F} = \ldots = c_{N,F}$



Optimal Sharing Strategy with Homogeneous QoS

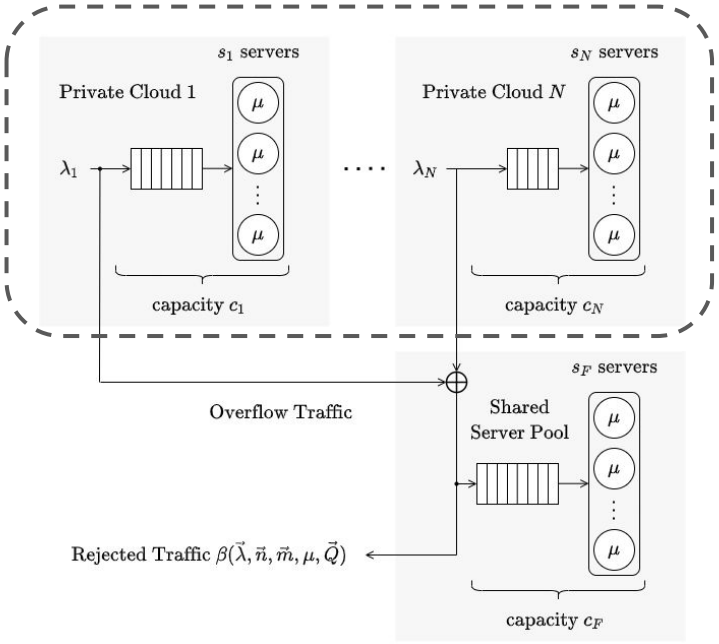


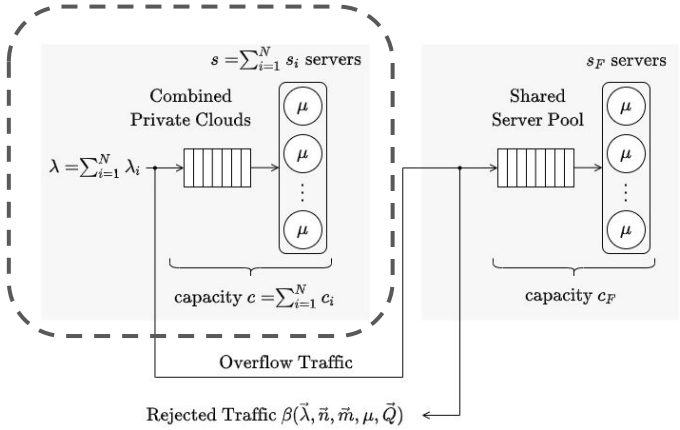


Given $c_i = \lfloor \mu(1+Q_i)s_i \rfloor$ and $c_{i,F} = \lfloor \mu(1+Q_i)\sum_{i=1}^{N} m_i \rfloor$, if $Q_i = Q$ and $m_i = n_i$, $\forall i \in \{1, ..., N\}$, then

- $c_{1,F} = c_{2,F} = \ldots = c_{N,F}$
- $c = \lfloor \mu(1+Q)(s_F + \sum_{i=1}^{N} s_i) \rfloor \geq c_1 + c_2 + \ldots + c_N + c_F$

Minimizing Rejected Traffic

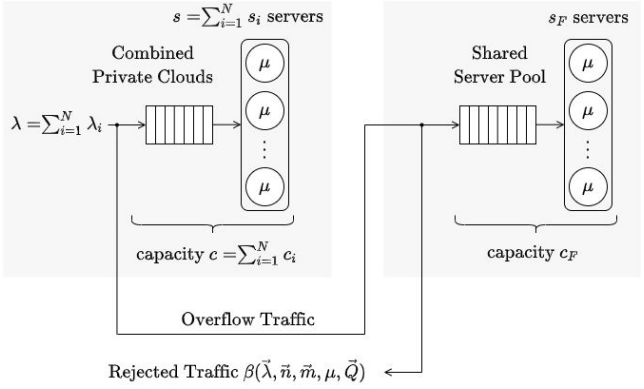


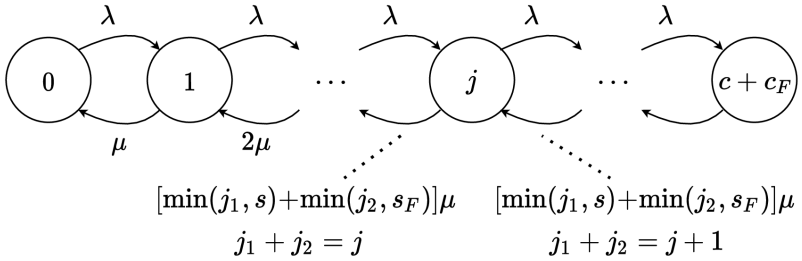


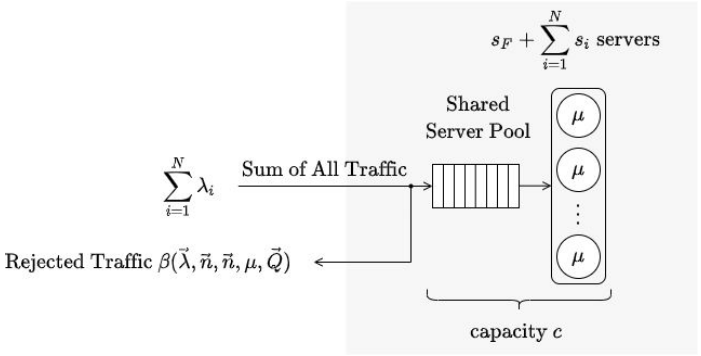
$$\sum_{i=1}^{N} \lambda_i B(n_i, c_i, \lambda_i/\mu) \geq \left(\sum_{j=1}^{N} \lambda_j \right) B\left(\sum_{j=1}^{N} n_j, \sum_{j=1}^{N} c_j, \sum_{j=1}^{N} \lambda_j/\mu \right)$$

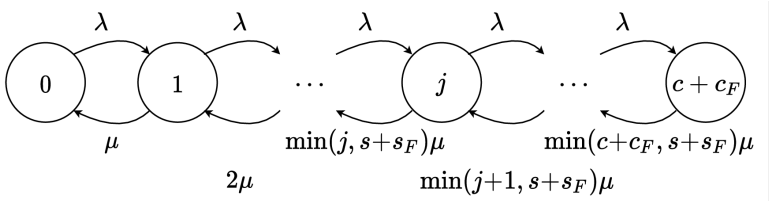
Smith, David R., and Ward Whitt. "Resource sharing for efficiency in traffic systems." *Bell System Technical Journal* 60.1 (1981): 39-55.

Minimizing Rejected Traffic

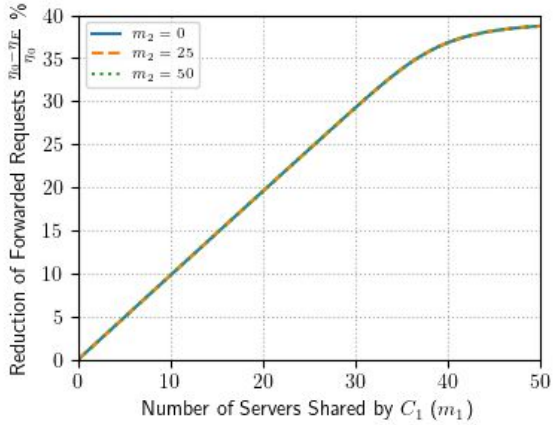




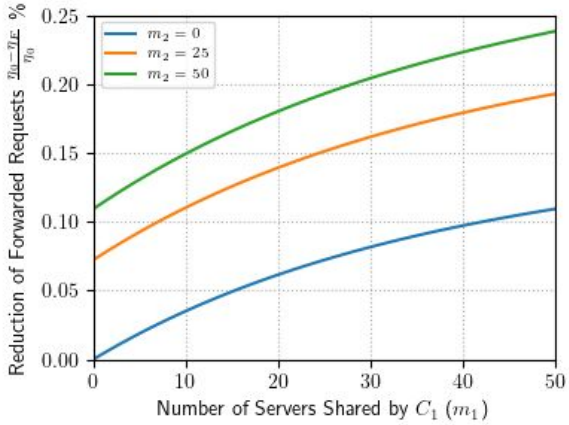




Sharing Strategy for Homogeneous QoS

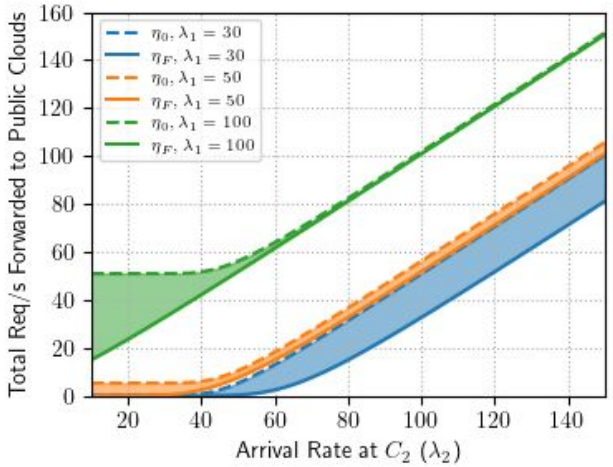


Given an over-provisioned (C1) and an under-provisioned (C2) clouds, sharing more C2 does not reduce rejected traffic (all of its resources are in use).



With 2 equally overloaded clouds, sharing either can help with the federation.

Performance (Sharing Everything)

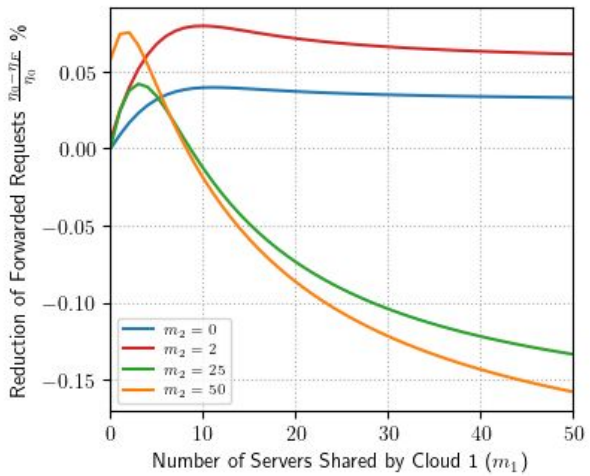


Best to group under-provisioned and over-provisioned clouds together when Q1=Q2.

(λ_1, λ_2)	η_0	η_F	δ_1	δ_2
$(30, 150)$	100.5	81.2	27.6%	7.1%
$(50, 150)$	105.7	101.0	5.9%	1.8%
$(100, 150)$	151.4	150.7	0.4%	0.3%

Table I: Forwarding rate to public clouds before (η_0) and after (η_F) joining the federation and relative cost reduction (δ_1, δ_2) for different arrival rates when $\vec{n} = \vec{m} = (50, 50)$, $\vec{Q} = (0, 0)$.

Heterogeneous QoS (Sharing Everything)

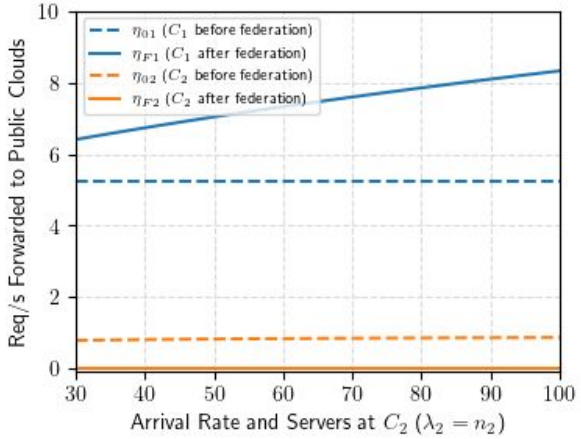


When Q1 = 0 and Q2 = 1, sharing more Cloud 2 servers causes more overflow from 2 and pushes 1's traffic out of federation more than 2 let in.

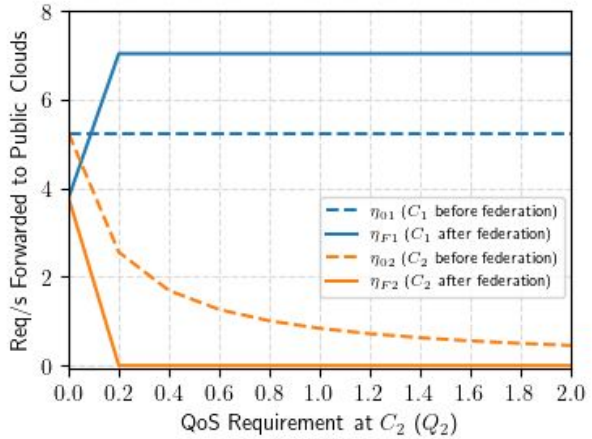
(λ_1, λ_2)	(n_1, n_2)	(Q_1, Q_2)	δ_1	δ_2
$(50, 100)$	$(50, 100)$	$(0, 10)$	-3.7%	-2.1%
$(50, 100)$	$(50, 100)$	$(0, 50)$	-3.8%	-2.2%
$(50, 1000)$	$(50, 1000)$	$(0, 50)$	-14.8%	-0.8%
$(7, 10000)$	$(10, 10000)$	$(0, 1000)$	-37.4%	-0.05%

Table II: Relative cost reduction with heterogeneous QoS

Heterogeneous QoS



Given Q1 = 0 and Q2 = 1, increasing Cloud 2's traffic and servers pushes more traffic from 1 to the public cloud.



With Q1 = 0, increasing Q2 quickly allows Cloud 2 to send 0 traffic to the public cloud, pushing traffic from 1 out of the federation but no further.

Profit Sharing

- Cloud Profit (CP) = revenue (constant) - operating cost (constant) - public cloud cost
- ΔCP = public cloud cost **before** federation - public cloud cost **after** federation
- Federation Profit = $R(S) = P\Big(\eta_0(S) - \eta_F(S)\Big) = \sum_{i=1}^{S} \Delta CP_i$

Reward policies:

1. $R_i^{SV}(S) := \sum_{S' \subseteq S \setminus \{i\}} \frac{|S'|!(|S|-|S'|-1)!}{|S|!} \big(R(S' \cup \{i\}) - R(S')\big)$

   Change in the federation profit

2. $R_i^{SR}(S) := \frac{m_i}{\sum_{j=1}^{K} m_j} R(S)$

   Shared resources

3. $R_i^{SIR}(S) := \frac{m_i(1-\rho_i)}{\sum_{j=1}^{K} m_j(1-\rho_j)} R(S)$

   Shared idle resources

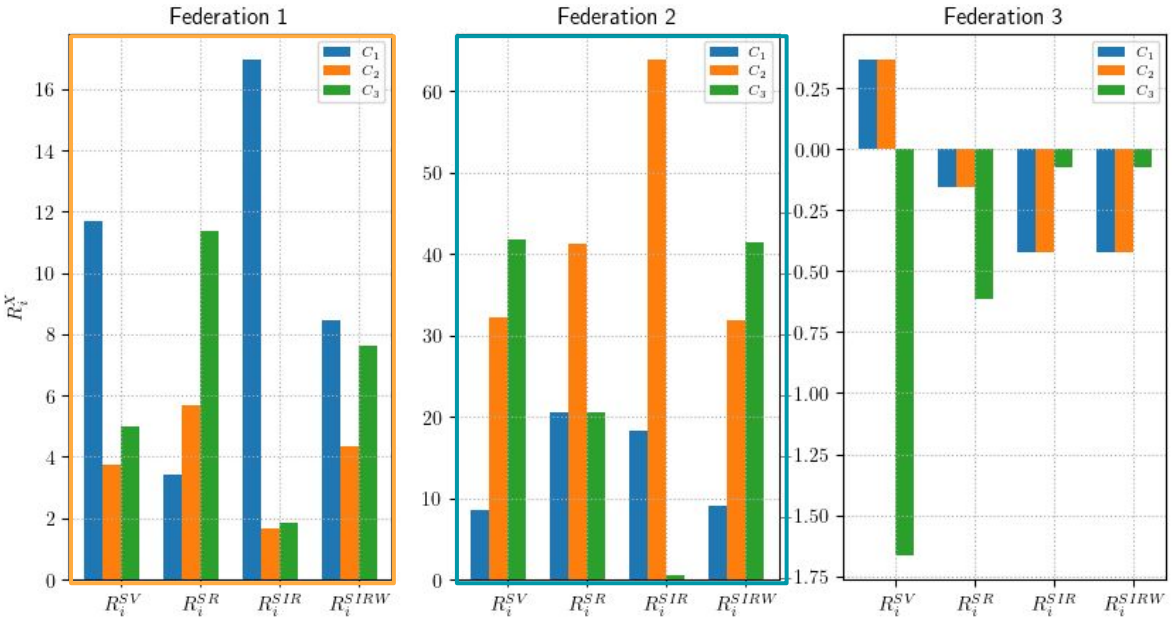
4. $R_i^{SIRW}(S) := \frac{1}{2}\left(R_i^{SIR}(S) + \frac{\lambda_i - \rho_i n_i \mu}{\sum_{j=1}^{K} \lambda_j - \rho_j n_j \mu} R(S)\right)$

   Both in shared idle resources and traffic overflow to the federation

Profit Sharing

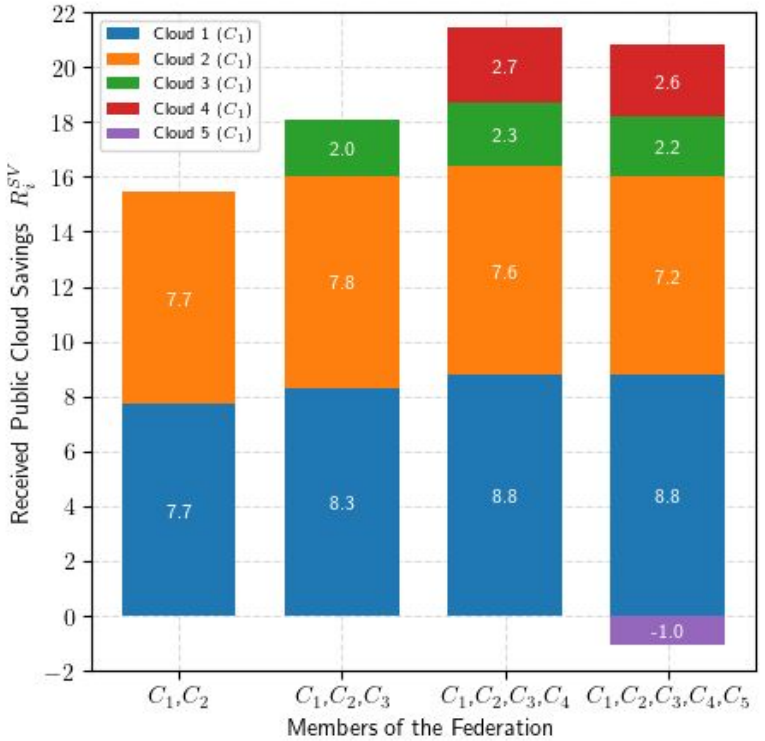
Cloud	Arrivals	Servers	Shared Servers	QoS
1/2/3	10/70/140	30/50/100	30/50/100	0/0/0
1/2/3	30/30/150	50/100/50	50/100/50	0/0/0
1/2/3	50/50/200	50/50/200	50/50/200	0/0/1

Contribution	Federation 1	Federation 2	Federation 3
Shared Resources	$C_3 > C_2 > C_1$	$C_2 > C_1 = C_3$	$C_3 > C_1 = C_2$
Shared Idle Resources	$C_1 > C_3 > C_2$	$C_2 > C_1 > C_3$	$C_1 = C_2 > C_3$
Overflow Traffic	$C_3 > C_2 > C_1$	$C_3 > C_1 > C_2$	$C_1 = C_2 > C_3$



Adding/Discouraging Members

Cloud	Arrivals	Servers	QoS
1	10	30	0
2	50	30	0
3	30	30	0
4	50	50	0
5	50	50	1



Summary of Contributions

- Evaluate effects of sharing strategies in performance of federations.
- Prove that clouds with the same QoS can maximize federation performance and profit by sharing everything.
- Propose profit sharing mechanism using public cloud saving to compute Shapley Values and illustrate ability to discourage “free riders”.

Future Work

- Heterogeneous workloads and resources
- Different distributions of workload sizes and interarrival times
- Define more stringent or dynamic queueing capacity to closely respect the exact QoS